LongevityBench-Mouse

A mouse-genetics benchmark for testing whether aging LLMs reason from phenotype evidence or lean on memorized gene names. Live dashboard: recall-vs-reason.lovable.app · Code: github.com/AndersonsRepo/longevity-redflag-bench

Dataset	Main eval	Single-gene control	Extra credit
74,573 MGI mouse genotypes	**240** controlled binary prompts (120 matched pairs)	**120** matched pairs = 240 prompts; epistasis removed	**12** reasoning stress-test prompts

Problem

Aging-biology LLMs can look good when an eval rewards recall of famous genes. Track 01 asks for verifiable benchmark tasks; our question was: **can a model infer a mouse mutation's lifespan effect from genotype, zygosity, and phenotype profile?**

Solution

We built ChatML/JSONL tasks from mouse genotype-phenotype annotations. Mortality/aging MP terms become the label; all non-mortality phenotype terms become the prompt. Each item is rendered twice: gene shown and gene hidden. The difference, `Delta_recall`, measures how much the model benefits from seeing the gene name.

Ground Truth

Labels are deterministic and auditable. MGI curators assign phenotype terms from published papers; MP ontology defines the mortality/aging subtree; we corrected bidirectional label bugs before scoring. No model-generated labels are used.

Data Sources

- **MGI** `MGI_PhenoGenoMP.rpt`: genotype, allele, zygosity, MP terms, PMIDs.
- **MP ontology** `mp.obo`: survival label boundary and phenotype-system categories.
- **GenAge/HAGR**: famous-gene blocklist for contamination control.
- **SynergyAge/OpenGenes**: explored for quantitative lifespan, but too sparse for the main task.

Tech Stack

Python, pandas, Pydantic, OpenAI-compatible Longevity-LLM endpoint, Claude Sonnet 4.6, Claude Haiku judge, Typst, Lovable/TanStack/Vite/React dashboard, static JSON feeds.

Team

Anderson Edmond, Ibrahim, CS teammate, biology teammates.

Tests We Built

- **LB-0138 binary survival:** Does this genotype impair survival? Controlled adult/aging curation across 12 phenotype systems; balanced yes/no; gene-shown vs gene-hidden matched pairs.
- **Random baseline:** Same binary format without curation, proving why random MGI sampling hides the effect.
- **LB-0150 pairwise extension:** Which mutation is more likely to extend lifespan?
- **LB-0154 ternary:** Shortens / no-effect / extends, added after scientist feedback so the model can choose neutral.

Extra Credit

- **Reasoning trace scorer** (`judge/score_trace.py`): checks gene hallucination, think/answer consistency, and pathway/system grounding against MGI-derived facts; optional Claude Haiku biological verification.
- **Live/demo API** (`api_server.py`, `judge/live_scorer.py`): streams Longevity-LLM reasoning plus scorer output for the frontend.
- **12-prompt stress test** (`testing/run_new_prompts.py`): multi-mutant, synthetic-gene, reverse-lookup, and gene-complement prompts. Result: 9/12 accuracy, mean trace score 0.682.

Recall reliance, controlled set	...collapses, single-gene only	Life-extension blind spot
+0.100 Longevity-LLM Delta_recall; McNemar p=0.017 (significant)	**+0.017** not significant (p=0.86) - the effect was a multi-gene/epistasis artifact	**1/20** Longevity extends-recall gene-shown vs 10/20 gene-hidden - showing the gene makes it WORSE

Key Findings

- **Gene-name reliance was mostly an epistasis artifact.** The mixed controlled set showed significant `Delta_recall`, but after removing multi-gene/transgene cases it collapsed to non-significant.
- **The life-extension blind spot persisted.** In ternary tests, both models recognized shortens/no-effect far better than extends; they default toward "mutation is harmful or neutral."
- **The 9B specialist tied frontier on binary survival.** Longevity-LLM was statistically indistinguishable from Claude Sonnet 4.6 on the controlled binary task.
- **Famous genes are memorized.** Gene-only contamination probes showed famous >> obscure recall, validating the obscure-gene design.

Why It Meets Track 01

- Preferred formats: binary, ternary, pairwise MCQ.
- Verifiable ground truth: MGI + PMIDs + MP ontology.
- Leakage control: train/test split by gene components; single-gene control for epistasis.
- Formal metrics: accuracy, macro-F1, McNemar exact tests, paired bootstrap CIs, `Delta_recall`.
- No sequence modality and no free-text expert scoring required for final answers.

Primary artifacts: `docs/LABELING.md`, `docs/EVALS.md`, `results/stats.json`, `results/single_gene/comparison.md`, `results/ternary_results.md`, `testing/new_prompts_results.json`, `judge/score_trace.py`, and the live Lovable dashboard (recall-vs-reason.lovable.app) backed by `results/dashboard_data.json`. Full code + data: github.com/AndersonsRepo/longevity-redflag-bench.